SESSION 1

GitHub Repository Setup

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	GitHub Repository Setup
Industry Mapping	Collaborative Development
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Team Members	Pawni Agrahari
Repository	[Add GitHub repository link]

Topics Covered in This Session

- Introduction to software engineering tools
- Introduction to Git and GitHub
- Repository creation and version control basics
- Software project lifecycle in industry
- Team formation and project domain selection

1. Project Overview

AI Career Guidance & Skill Recommendation System is a web-based platform that analyses a student's resume, skills, education, projects and interests, and recommends suitable career paths. It identifies skill gaps, generates a personalised learning roadmap, suggests courses, shows analytics, and offers an AI career chatbot powered by the Claude API.

Item	Detail
Chosen domain	EdTech / Career Guidance with Applied AI
Tech stack	React (frontend), Node.js + Express (backend), MongoDB (database), Claude API (AI), Python + Pandas (analytics)
Version control	Git (local) and GitHub (remote, collaboration, issues, pull requests)
Suggested repo name	ai-career-guidance-system

2. Tools Introduced

Tool	Purpose in this project
Git	Distributed version control: tracks every change, supports branching and merging.
GitHub	Remote hosting, pull requests, code review, issue tracking, project board, documentation.
VS Code	Editor with built-in Git support, ESLint and Prettier extensions.
Postman	API testing for Express routes during development.
MongoDB Compass	Visual inspection of collections and documents.
Draw.io / StarUML	UML diagrams (Sessions 5 to 7).
Google Docs / Overleaf	SRS and report documentation (Session 4).

3. Version Control Basics

Git stores a project as a series of **commits** (snapshots). Each developer works on a local copy (clone) and shares work through a remote repository.

Concept	Meaning
Repository	The project folder tracked by Git, including its full history.
Commit	A saved snapshot with an author, timestamp and message.
Branch	An independent line of development, used to build features safely.
Merge / Pull Request	Combining a branch into another after review.
Remote / origin	The GitHub copy of the repository.
Clone, Fork	Clone copies a repo locally; fork copies it to your own GitHub account.
.gitignore	Lists files Git must not track (node_modules, .env, build output).

4. Repository Setup Steps

4.1 Create and push the repository

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
mkdir ai-career-guidance-system && cd ai-career-guidance-system
git init
echo "# AI Career Guidance System" > README.md
git add .
git commit -m "chore: initial commit"
git branch -M main
git remote add origin https://github.com/<username>/ai-career-guidance-system.git
git push -u origin main
```

4.2 Daily workflow commands

Command	What it does
git status	Shows changed and untracked files.
git add <file>	Stages changes for the next commit.
git commit -m "msg"	Records staged changes.
git pull origin develop	Fetches and merges the latest team changes.
git checkout -b feature/resume-upload	Creates and switches to a new branch.
git push origin <branch>	Uploads the branch to GitHub.
git log --oneline --graph	Shows compact commit history.

5. Proposed Repository Structure

```
ai-career-guidance-system/
  client/                 React frontend (pages, components, services)
  server/                 Express backend (routes, controllers, models, services)
  analytics/              Python + Pandas scripts for dashboard data
  docs/                   Session deliverables (PDFs) and diagrams
    session-01 ... session-09/
  .github/                Issue and pull request templates
  .gitignore   README.md   LICENSE
```

6. Branching Strategy and Conventions

Branch	Purpose	Rule
main	Stable, demo-ready code	Protected; merge only through reviewed pull requests
develop	Integration branch	Features merge here first
feature/<name>	One feature or task	Branch from develop; delete after merge
fix/<name>	Bug fixes	Same flow as feature branches
docs/<name>	Documentation work	Used for SRS, UML and report updates

Commit message convention (Conventional Commits): `feat:` new feature, `fix:` bug fix, `docs:` documentation, `refactor:` code restructuring, `chore:` tooling and setup. Example: `feat: add resume upload endpoint`.

- Every pull request needs at least one teammate review before merging.
- Never commit secrets: API keys live in a `.env` file listed in .gitignore.
- Use GitHub Issues for tasks and a Projects board (To Do, In Progress, Review, Done) to track progress.

7. Software Project Lifecycle in Industry

Industry teams follow a lifecycle (SDLC) so that work moves from an idea to a maintained product in controlled stages. Our course sessions mirror these stages.

Figure 1: Software development lifecycle

Stage	Industry activity	Course session
Plan / Analyse	Problem definition, feasibility, requirement gathering, SRS	Sessions 2, 3, 4
Design	UML models, design principles, architecture and patterns	Sessions 5 to 9
Develop	Feature branches, code review, continuous integration	Git and GitHub workflow (this session)
Test and Deploy	Unit and integration tests, staging, release	Later sessions
Maintain	Bug fixes, monitoring, new features from user feedback	Issues and releases on GitHub

Process model: The team will follow an **Agile (Scrum-style)** approach with short sprints, because requirements for an AI product evolve as the team learns what users value. Each sprint produces a working increment that is merged into develop.

8. Team Formation

Role	Responsibilities	Member
Project Lead / Scrum Master	Planning, sprint tracking, merging to main	Pawni Agrahari
Frontend Developer	React UI, dashboard charts, chatbot interface	Pawni Agrahari
Backend Developer	Express APIs, authentication, MongoDB models	Pawni Agrahari
AI / Data Engineer	Claude prompts, resume parsing, Python + Pandas analytics	Pawni Agrahari
Documentation and QA	SRS, UML diagrams, test cases, README	Pawni Agrahari

Roles can be shared in smaller teams; every member must own at least one module and commit regularly.

9. Project Domain Selection

Criterion	Why this domain fits
Real-world need	Students often lack clear guidance on which career fits their skills and what to learn next.
Use of modern technology	Combines full stack web development, generative AI and data analytics.
Feasibility	All tools have free tiers or student plans; scope can be delivered in a semester.
Scalability	Can grow into mentoring, job matching and placement-cell analytics.

10. Deliverable Checklist

Item	Status
GitHub repository created with README and .gitignore	Done
Branches main and develop created; main protected	Done
Team members added as collaborators	To do
Issue board with initial tasks	To do
docs/session-01 folder holds this PDF	Done